

Machine-Learning-Guided Design of Thermostable PET-Degrading Enzymes via a Stacked Multi-Model $\Delta\Delta G$ Ensemble

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Abstract for Poster Presentation · Computational Bioengineering / Enzyme Design

19,071 EXPERIMENTAL MUTATIONS	73.1% HELD-OUT ACCURACY	70.6% SIGN ACCURACY (S669)	1.10 MAE KCAL/MOL
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BACKGROUND & MOTIVATION

Polyethylene terephthalate (PET) is one of the most abundant plastic pollutants, and enzymatic depolymerization by PETases and cutinases offers a sustainable recycling route. The central barrier to industrial deployment is enzyme thermostability: wild-type *Ideonella sakaiensis* PETase denatures near 40 °C, far below the ~70 °C window where PET becomes most accessible to catalysis. Rational stabilization requires accurately predicting how individual point mutations perturb folding free energy ($\Delta\Delta G$) — a long-standing, unsolved challenge at the interface of biophysics, structural biology, and machine learning.

APPROACH

We developed an end-to-end, machine-learning-guided design platform that predicts mutation-induced $\Delta\Delta G$ and melting-temperature shift, ranks candidate variants, and renders mutations on experimentally accurate 3-D structures. A **stacked ensemble of eight regressors** (Gradient Boosting, XGBoost, LightGBM, CatBoost, Histogram Gradient Boosting, a neural network, Random Forest, and Extra Trees) was trained on **19,071 experimentally measured mutations** from FireProtDB, ProDDG, and ThermoMutDB — with **no synthetic data**. Each mutation is encoded by **144 features** spanning physicochemical deltas, relative solvent accessibility and secondary structure, PSSM evolutionary conservation, ESM-2 and ESM-1v protein-language-model representations, metal-coordination geometry, Debye–Hückel/Born electrostatic physics, assay temperature and pH, and AlphaFold per-residue confidence. Generalization was assessed on the **S669 benchmark** — 669 mutations from proteins entirely absent from training — with all overlapping entries removed for strict independence.

RESULTS & SIGNIFICANCE

On 10-fold cross-validation the ensemble reached 79.7% stability-class accuracy (Pearson $r = 0.76$; MAE = 0.92 kcal/mol). On the fully independent S669 benchmark it achieved **73.1% accuracy and 70.6% sign accuracy** — correctly calling the direction of stability change for 7 of every 10 unseen mutations — with MAE = 1.10 kcal/mol and Pearson $r = 0.39$ – 0.41 , competitive with established $\Delta\Delta G$ predictors (literature S669 correlations typically 0.4–0.55) and exceeding classical force-field tools such

as FoldX. A companion ΔT_m regressor attained MAE = 3.66 °C (Pearson r = 0.65). The ~6.6-point cross-validation-to-test gap is reported transparently, quantifying the genuine difficulty of extrapolating to novel protein families. Integrated into an interactive design tool, the model proposes and visualizes stabilizing PETase variants on resolved structures, providing a practical, honestly benchmarked route toward engineering thermostable plastic-degrading enzymes for sustainable PET recycling.

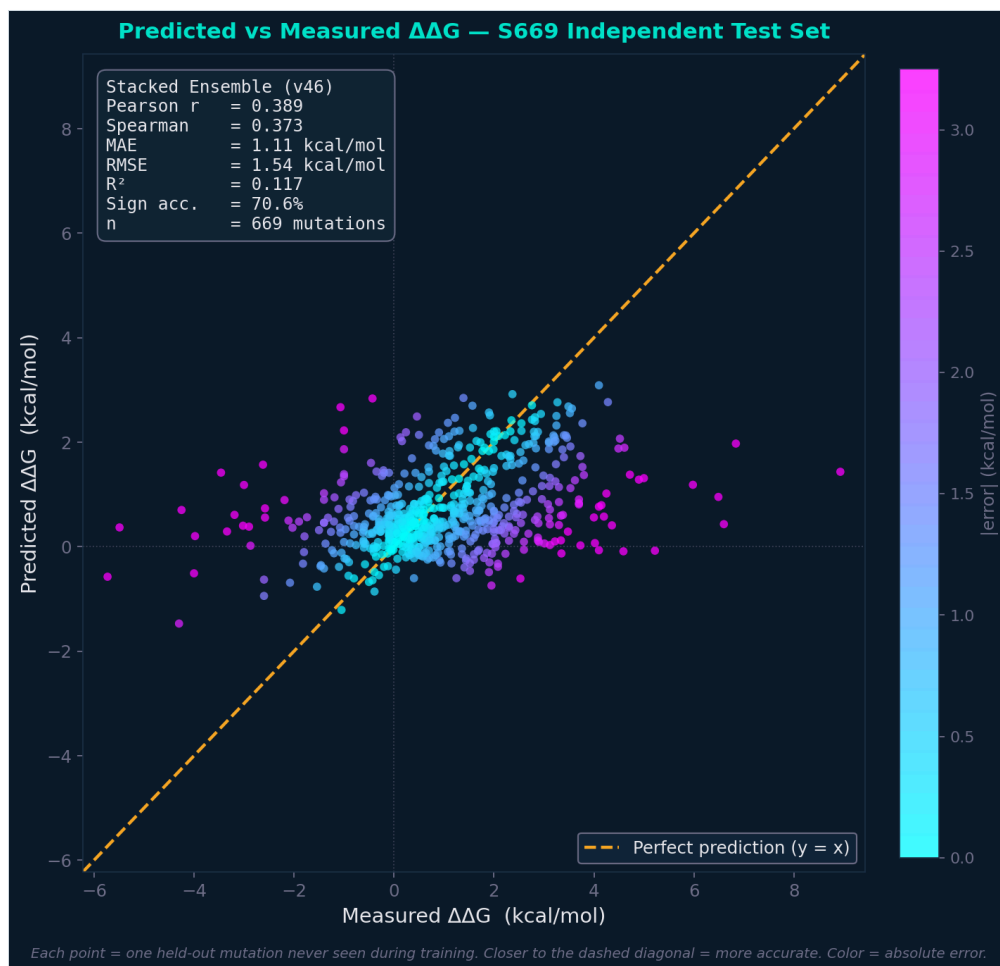


Figure 1. Predicted vs. measured $\Delta\Delta G$ on the S669 independent test set (n = 669 held-out mutations). Dashed line = perfect prediction; color = absolute error. Pearson r = 0.39, MAE = 1.10 kcal/mol, 70.6% sign accuracy.